LORIS

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Open Source Data Management for Multi-Site Research

Accelerating reproducible research for 25+ years

125,000+ PARTICIPANTS

300,000+ SCANS

260,000+ DATA VARIABLES

400+ RESEARCH SITES

LORIS is a **web-based, open-source data and project management platform** for biomedical research. It enables research teams to store, cross-link, curate, visualize, and share multi-modal data — including neuroimaging, behavioural, clinical, electrophysiological, biospecimen, and genetic data — securely over the web. LORIS is developed and maintained by the **McGill Centre for Integrative Neuroscience (MCIN)** at the Montreal Neurological Institute-Hospital, led by **Alan Evans and Samir Das**. It is freely available under the GPL v3 licence.

THE CHALLENGE

Managing Multi-Site Longitudinal Neuroscience Data

Large-scale neuroimaging studies span multiple sites, time points, and data modalities. Without standardised infrastructure, data becomes fragmented, quality control inconsistent, and sharing nearly impossible. Subjects drop out; protocols drift; formats diverge.

LORIS was designed from the ground up to address these challenges — automating data flow from acquisition and quality control through processing and open dissemination, with built-in FAIR data principles.

LORIS ENABLES

- ✦ Multi-site, longitudinal data entry & management
- ✦ Automated imaging QC and protocol validation
- ✦ Flexible instrument builder for clinical measures
- ✦ RESTful API for integration with external pipelines
- ✦ Granular, role-based access and audit trails
- ✦ FAIR-aligned data sharing and dissemination

📊 Data Acquisition

Tablet-friendly data entry, scanner-to-LORIS imaging pipelines (DICOM), survey modules for remote participant entry, and multi-modal integration from day one.

🔍 Quality Control

Automated protocol compliance checking, double data entry conflict detection, visual MRI QC, in-browser 3D BrainBrowser visualisation, and the Analytics Dashboard.

🔗 Analysis & Sharing

NoSQL Data Querying Tool for cross-modal export, RESTful API, CBRAIN HPC integration, BIDS neuroimaging standard support, and open data release capability.

Open source (GPL v3) · loris.ca · github.com/aces/Loris · demo.loris.ca ·
Developed at MCIN, Montreal Neurological Institute

LORIS Feature Suite — A Complete Research Data Lifecycle Platform

01 · Candidate & Visit Management

Subject-centric organisation tracking all longitudinal visits, timepoints, and cohort membership. Supports double data entry with automatic conflict detection, custom visit scheduling, and tablet-friendly forms for clinical assessors.

02 · Imaging-EEG Pipeline & MRI QC

Automated multi-modal ingestion, protocol compliance checking, and in-browser 3D image QC via **BrainBrowser**. Supports MRI, fMRI, PET, EEG, MEG. Integrates with LORIS-MRI for imaging insertion and QC workflows, including incidental findings tracking.

03 · Instrument Builder & Behavioural Data

Non-technical staff can build clinical and neuropsychological instruments via the Instrument Builder. Custom scoring can be implemented in a variety of languages. At-home Surveys support secure data entry by participants and include automated email reminders.

04 · Data Querying Tool (DQT)

A cross-modal querying tool enabling granular filtering across all data types — **neuroimaging, electrophysiological, behavioural, genetic, and biospecimen** — with CSV/JSON export to downstream analysis pipelines and CBRAIN HPC.

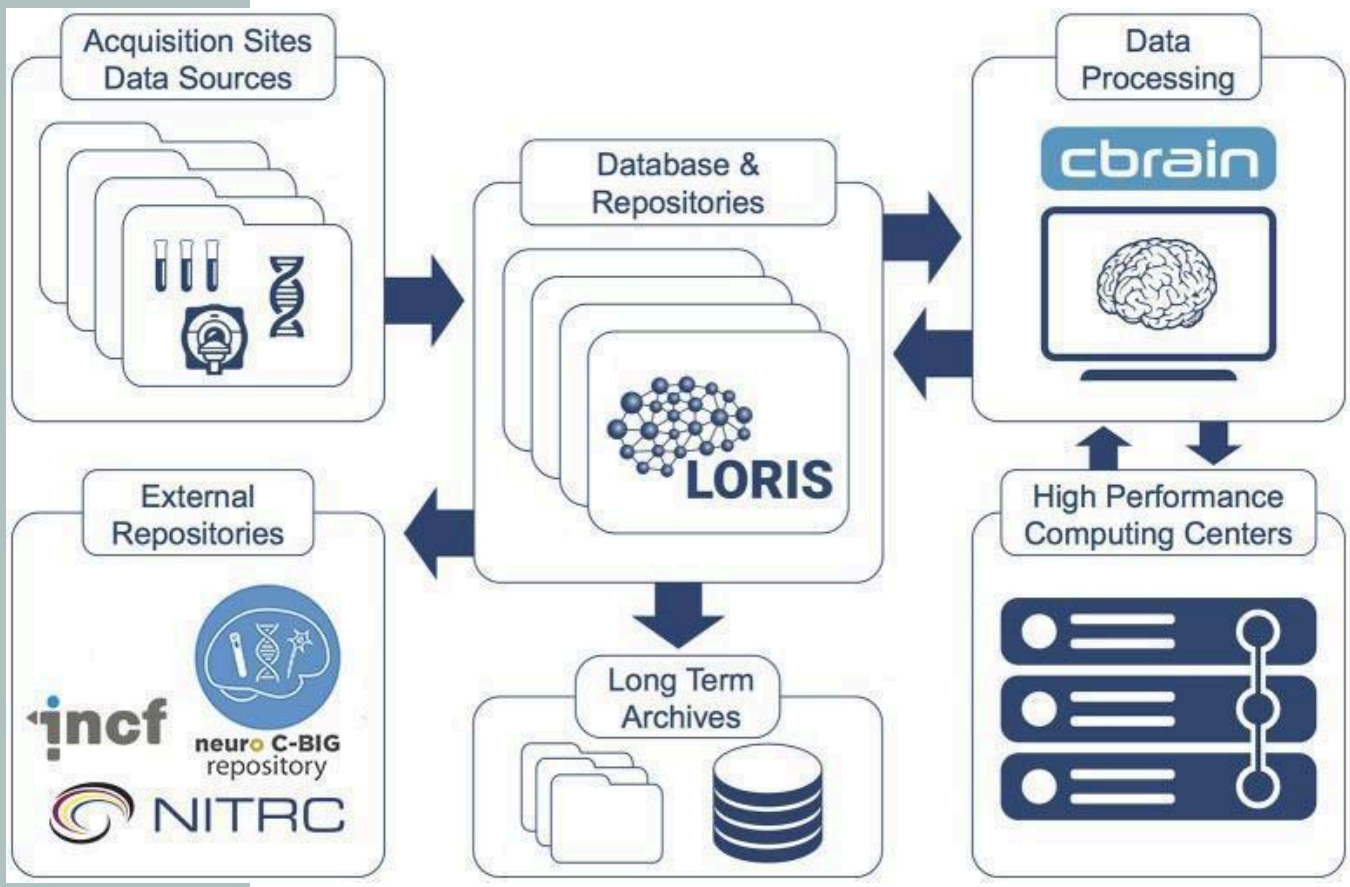
05 · Biorepository & Genomics

Integrated biospecimen tracking (sample collection, storage, processing) with data sharing capabilities. The Genomic Browser integrates summary genomic data alongside neuroimaging phenotypes.

06 · Security, Access Control & Audit Tools

Granular role-based access control at site, module, workflow, and data-modality levels. Privacy-by-design architecture with full audit trails, encrypted transfer and storage, and LORIS-wide session management.

Technology stack: PHP · MySQL · React/JSX · Apache · RESTful API · BrainBrowser (WebGL)
BIDS compliant · CBRAIN (HPC) · Docker · GPL v3



Deployments & Use Cases — Global Research Community

400+ research sites worldwide

Self-Supported Deployments

Install LORIS on self-hosted web servers. MCIN provides documentation, a mailing list, and active GitHub issue support.

- CCNA: Canadian Consortium for Neurodegeneration in Aging
- Infant Brain Imaging Study on Autism (IBIS)
- NKI-Rockland sample : Child Mind Institute, New York: Milham Research Group
- HBCD: HEALthy Brain and Child Development Study
- Q1K Autism Platform, 2020–
- NIH MRI Study of Normal Brain Development (NIHPD), 1999–
- NeuroDevNet (KBHN) Imaging Subprojects
- UK BioBank Data Node, 2020–2026
- 1000 Brains (1000-Gehirne) Project, Germany
- KISTI, Korea: Neurodegeneration Research Group
- GUSTO, Singapore: Population Data from National Birth Cohorts
- PERFORM Centre, Concordia University
- Canadian Neonatal Brain Platform
- CALSNIC: Canadian ALS Neuroimaging Consortium
- Douglas Mental Health Institute — Brain Imaging Centre
- Leukodystrophy MNI-based Databank
- Cuban Neuroscience Centre: CBHMP
- University of Edinburgh, Centre for Clinical Brain Sciences

Open Science Deployments

LORIS supports institutional open data sharing policies, providing data portals aligned with FAIR principles.

ABIDE: Autism Brain Imaging Data Exchange
BigBrain: High-resolution 3D histological model of human brain (HIBALL/Jülich)
CCNA: Canadian Consortium for Neurodegeneration in Aging (COMPASS-ND)
PREVENT-AD: Open database on the Canadian Open Neuroscience Platform (CONP)
MNI Open iEEG Atlas: Sleep/wake atlas of normal brain activity
TOSI & C-BIGR: Open Science Biobank at Montreal Neurological Institute-Hospital
QPN: Quebec Parkinson's Network
EEGNet: EEG data hub (eegnet.loris.ca)

Licensing, Community & Publications

Licensing & Installation

Freely downloadable from GitHub under GPL v3. Runs on any web-accessible Unix server (Ubuntu or CentOS). Try it first via the live demo at **demo.loris.ca**.

Community & Support

Active developer mailing list, GitHub Issues, and a contributing community spanning MCIN and affiliated institutions globally. 20+ active contributors committing ≥25% time weekly to core LORIS development.

Key Publications

Das, LORIS, 2012
Das, MNI Ecosystem, 2016
Das, Cyberinfrastructure. 2017
Das, C-BIG Repository, 2022

Architecture, Roadmap & Getting Started

System Architecture

Data Acquisition Layer	Processing & Quality Control	Sharing & Dissemination
MRI/EEG scanner → LORIS-MRI pipeline → DICOM ingestion Instrument Builder for behavioural/clinical forms Tablet data entry + remote surveys Biospecimen tracking & results sharing	Automated QC protocol compliance BrainBrowser 3D in-browser MRI visualisation (WebGL) Dashboard at-a-glance visualization CBRAIN HPC integration for pipeline execution	Data Querying: cross-modal queries RESTful API for automated integration BIDS neuroimaging standard export Configurable tiered access data portal

Strategic Roadmap

i | BIDS Integration & Open Science Standards

Full BIDS (Brain Imaging Data Structure) compliant interoperability for LORIS datasets — enabling seamless interoperability with BIDS-compatible analysis tools and open data repositories such as OpenNeuro and the CONP Portal.

ii | Enhanced RESTful API & External Pipeline Integration

Ongoing expansion of the LORIS RESTful API to cover all modules, enabling richer programmatic access. Integration with CBRAIN for HPC-based processing and with the CONP Portal for federated dataset discovery.

iii | Privacy-by-Design & Consent Code Frameworks

Expanding LORIS's consent code implementation to align with international data-sharing standards (GA4GH, GDPR). Enabling granular participant consent tracking across open and restricted data portals.

iv | Modular Expansion: Imaging, Genetics, EEG, Biospecimens

Ongoing development of specialised modules — including Genomic Browser (summary genomic data), EEG/MEG data management, and biorepository management — maintaining LORIS as a truly multi-modal platform.

Getting Started & Contact

LORIS

Website & Demo
loris.ca
demo.loris.ca

Code & Docs
github.com/aces/Loris
loris.ca/getting-started

Funders & Acknowledgements
Brain Canada · NIH · CIHR · CFREF
CFI · Healthy Brains Healthy Lives (HBHL)
Ludmer Centre for Neuroinformatics and Mental Health · Fonds de Recherche du Québec · Montreal Neurological Institute-Hospital · Tanenbaum Open Science Institute (TOSI)

Core Team & Leadership
PI: Alan Evans, MCIN/MNI, McGill University
Associate Director: Samir Das
20+ core developers at MCIN and affiliated institutions
6000+ merged pull requests · 196 forks
13,000+ commits · GPL v3

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